



DONATUS OMONDI
ELECTRICAL AND
ELECTRONICS ENGINEER

CONTACT INFO

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Nairobi, Kenya

EDUCATION

Sep 2019 - June 2024
Bachelor's Degree Electrical and
Electronics Engineering
University of Nairobi
Award: Second Class Honours
Upper Division

2015 - 2018
Maranda High School
Grade: A-

2007 - 2014
St. Hannahs Preparatory School
Grade: A

LANGUAGES

English
Swahili

REFEREE

oscaror@uonbi.ac.ke
[Website](#)

PROFILE

Embedded Systems Engineer with 2+ years of hands-on experience in firmware development, hardware-software integration, and PCB design for production and prototype systems. Strong background in STM32 and ESP32 platforms, real-time systems using FreeRTOS, and communication-heavy applications including CAN, Modbus, MQTT, and OCPP. Demonstrated ability to identify and correct critical design flaws, implement scalable telemetry pipelines, and deliver robust embedded solutions in EV charging, IoT tracking, and smart agriculture domains.

EXPERIENCE

FEB 2025 - PRESENT
Geviton Enterprises Ltd.
Full-time

EMBEDDED HARDWARE ENGINEER

- Designed and implemented BMS and charger drivers for STM32- and ESP32-based systems, enabling reliable EV battery charging and monitoring.
- Implemented OCPP-compliant firmware for EV charging infrastructure, supporting automated charging sessions and billing workflows.
- Developed telemetry and data streaming pipelines over HTTP for remote monitoring and diagnostics.
- Identified and corrected critical PCB design flaws prior to manufacturing, preventing costly board respins and resolving long-standing hardware issues.
- Developed Python packages containing custom C extensions to meet performance and portability requirements across embedded and Linux-based systems.

JUL 2024 - JAN 2025
Geviton Enterprises Ltd.
Full-time

GRADUATE TRAINEE

- Developed GSM and gas sensor drivers, enabling reliable data acquisition and communication for field-deployed devices.
- Created Python-based test and certification scripts, significantly improving efficiency during mass device certification.
- Designed and implemented firmware for vehicle and motorbike trackers with remote immobilization ("kill switch") functionality.
- Contributed firmware for remote land tracking devices used in asset and boundary monitoring.
- Reviewed and validated up to 6-layer PCB designs, issuing critical corrections before fabrication.

OCT 2023 - MAY 2024
Geviton Enterprises Ltd.
Part-time

STUDENT INTERN EMBEDDED FIRMWARE ENGINEER

- Developed GSM, RTC, and LoRa drivers for smart farming and battery metering hardware.
- Implemented secure telemetry pipelines (MQTTs) for cloud-based data aggregation.
- Reviewed PCB designs and provided firmware-informed feedback to improve future hardware revisions.
- Extended MicroPython firmware with low-level C/C++ modules, significantly reducing power consumption and improving runtime efficiency.

JUNE 2023 - OCT 2023
Geviton Enterprises Ltd.
Full-time

STUDENT ON ATTACHMENT EMBEDDED SYSTEMS ENGINEER

Involved in development of hardware solutions. Responsibilities include pcb assembly in SMT line, assembly of final product in enclosures, firmware development for STM32, RP2040 and ESP8266 based boards

AUG 2022 - MAY 2023
Tafiti Nanosatellite
Part-time

STUDENT ON ATTACHMENT ELECTRICAL TEAM

Development of the ADCS Subsystem of the KSA-funded Tafiti 3U CubeSat

JUN 2022 - AUG 2022

Gearbox
Part-time

STUDENT ON ATTACHMENT

Created a flask server running on a raspberry pi on the local area network to handle user registration and image uploads. These images were used to train the AI module for face recognition access

CERTIFICATIONS

AUG 2022

HCIA Datacom

Huawei Certified ICT Associate Certificate

OTHER WORK

FEB 2023

Battery Charge
Control

Used Raspberry Pi Pico W to automate charging of a laptop through sub-net requests to flask server running in laptop to protect remote devices.

SEPT 2021 - AUG 2021

Fulfilment &
Expense Track

Created a web application to track deliveries, partial payments by customers to deliveries, expenses incurred by business and business intertransactions using Python, Flask and Javascript. This is currently being used by 3 businesses

JULY 2021

Flask-dt python
package

Published a package to PYPI to dynamically display table records on user defined routes.

PROFESSIONAL SKILLS

C



C++



PYTHON



KICAD



LINUX



OCPP



LITTLEFS



INTERESTS

</>Programming

🎮Gaming

⚙️Robotics